

VYAZ'MA, Russian Federation

WMO: 266950

Lat: 55.1570N

Lon: 34.3840E

Elev: 822

StdP: 14.26

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-5.2	-15.3	2.5	-10.5	-9.3	3.4	-4.4	21.6	28.5	18.8	27.8	2.3	350	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	83.3	67.5	80.0	66.0	76.8	64.6	69.6	80.3	67.7	77.1	66.1	74.6	6.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	98.3	74.0	64.4	93.4	72.4	62.8	88.2	70.1	34.1	80.3	32.6	77.1	31.2	74.8	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	18.7	16.3	14.2	DB	-19.6	86.8	6.1	4.3	-24.0	89.9	-27.5	92.4	-30.9	94.8	-35.3	97.9
(5)				WB	-19.9	71.7	6.0	2.3	-24.2	73.3	-27.7	74.6	-31.1	75.9	-35.4	77.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	19.0	19.5	28.7	43.0	53.8	60.6	65.2	61.4	51.7	41.5	30.8	22.9
	DBStd	18.53	11.53	12.20	8.65	8.67	8.56	6.40	5.68	6.17	6.66	7.46	9.39	11.42
	HDD50	4530	962	854	659	245	57	2	0	0	57	278	575	841
	HDD65	8680	1427	1274	1124	660	358	163	67	147	401	728	1025	1306
	CDD50	1479	0	0	0	35	176	319	471	355	108	15	0	0
	CDD65	153	0	0	0	0	12	30	73	36	2	0	0	0
	CDH74	1382	0	0	0	5	120	249	629	361	18	0	0	0
	CDH80	325	0	0	0	0	16	35	163	110	1	0	0	0
Wind	WSAvg	6.2	7.0	6.8	7.0	6.3	6.1	5.6	4.7	5.1	5.3	6.5	6.9	7.1
Precipitation	PrecAvg	25.8	1.7	1.4	1.4	1.3	2.6	2.9	3.3	3.2	2.1	2.3	1.9	1.7
	PrecMax	36.9	2.8	2.7	2.9	3.3	5.8	6.9	7.5	10.1	4.9	5.4	4.0	3.7
	PrecMin	17.1	0.6	0.5	0.4	0.2	0.9	0.5	0.4	0.3	0.4	0.2	0.0	0.6
	PrecStd	5.3	0.6	0.6	0.6	0.7	1.2	1.5	2.0	2.4	0.9	1.3	0.8	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	38.9	56.1	72.8	82.2	83.7	89.9	90.2	76.5	66.4	51.9	46.1
		MCWB	38.6	36.2	45.6	54.0	65.4	66.9	69.7	68.9	63.3	55.4	49.3	45.2
	2%	DB	36.1	36.6	48.3	67.4	77.6	80.3	84.1	82.2	71.2	59.0	47.3	41.3
		MCWB	34.8	34.5	39.9	52.1	62.7	66.2	68.1	67.2	60.1	53.8	45.4	40.2
	5%	DB	34.2	35.0	43.5	62.7	73.0	76.7	80.4	77.4	66.6	55.1	44.8	37.6
		MCWB	33.1	33.2	37.6	50.6	59.9	64.0	66.9	65.1	57.9	51.3	43.1	36.5
	10%	DB	32.6	33.8	39.6	58.3	68.8	73.1	76.9	73.6	62.9	51.9	42.2	34.9
		MCWB	31.7	32.4	35.4	48.1	57.1	62.4	65.7	63.3	56.0	48.8	40.9	33.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	37.2	45.9	55.9	67.8	69.9	72.5	70.9	65.5	57.7	50.1	45.4
		MCDB	39.3	38.4	55.1	68.1	79.5	79.7	83.2	82.2	72.1	62.3	51.6	46.2
	2%	WB	35.1	35.0	41.2	53.3	63.9	67.5	70.2	68.6	61.4	54.6	45.8	40.3
		MCDB	35.8	35.9	46.6	64.6	74.6	77.0	80.9	79.6	68.3	58.4	46.9	41.1
	5%	WB	33.3	33.7	38.3	51.2	61.5	65.6	68.3	66.7	58.9	51.6	43.2	36.7
		MCDB	34.0	34.5	42.2	60.6	70.6	74.5	77.4	75.7	64.7	54.2	44.5	37.5
	10%	WB	31.8	32.5	36.0	49.1	58.9	63.5	66.6	64.6	56.7	49.3	41.0	34.1
		MCDB	32.5	33.5	39.2	57.6	66.4	71.0	74.9	71.6	62.0	51.7	42.0	34.7
Mean Daily Temperature Range	5% DB	MDBR	8.6	11.5	13.2	16.8	18.4	16.9	17.6	17.5	16.0	11.2	7.5	7.6
		MCDBR	5.9	7.7	18.4	23.8	24.5	22.4	23.6	24.2	22.8	16.3	9.8	6.8
		MCWBR	6.1	6.7	11.5	11.7	11.5	10.4	9.8	10.7	12.6	10.7	8.4	6.7
	5% WB	MCDBR	6.1	6.7	15.5	20.8	22.1	19.8	20.4	21.9	19.5	13.7	8.4	6.7
		MCWBR	6.2	6.3	10.3	11.0	11.1	10.4	9.7	10.7	11.4	10.0	7.8	6.8
Clear-Sky Solar Irradiance	taub	0.282	0.318	0.339	0.404	0.390	0.371	0.404	0.418	0.368	0.345	0.318	0.282	
	taud	2.129	2.078	2.137	2.165	2.312	2.387	2.311	2.262	2.375	2.387	2.248	2.131	
	Ebn at Noon	198	232	259	258	271	278	265	252	250	227	184	168	
	Edh at Noon	20	30	36	41	38	35	37	37	29	23	18	16	
All-Sky Solar Radiation	RadAvg	203	484	965	1281	1676	1773	1714	1381	912	426	165	120	
	RadStd	43	65	98	109	131	162	170	154	106	77	33	34	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+1.21	N/A	N/A	N/A	N/A	N/A	-232	-362	+167	N/A

Nomenclature: See separate page